| Ex:No: 6B<br>Date: 08/04/26 | Iris Flower Species Classification Using Naive Bayes Algorithm |
|---|---|

**Problem Description:**

The Iris dataset contains measurements of iris flowers belonging to three species: **Setosa, Versicolor, and Virginica**. Each flower is described using four numerical attributes: **Sepal Length, Sepal Width, Petal Length, and Petal Width.**

The objective is to classify iris flowers into their respective species using the **Naive Bayes classifier** and visualize the data distribution and classification results.

**Objective:**

To implement the Naive Bayes classification algorithm in R for predicting iris flower species and to visualize the dataset and classification results.

**Procedure:**

- **Step 1: Load Required Libraries**

  We load the necessary libraries for implementing the Naïve Bayes algorithm.

```
# Install package
install packages("e1071")

# Load required libraries
library (e 1071)
library (caret)
library (ggplot 2)
```

- **Step 2: Load and Prepare the Dataset**

  We use the iris dataset for classification and check its structure.

```
# Load the iris dataset
data (iris)

# Display first few rows
head (iris)
```

> head(iris)

| | sepal.length | sepal.width | petal.length | petal.width | Species |
|---|---|---|---|---|---|
| 1 | 5.1 | 3.5 | 1.4 | 0.2 | Setosa |
| 2 | 4.9 | 3.0 | 1.4 | 0.2 | Setosa |
| 3 | 4.7 | 3.2 | 1.3 | 0.2 | Setosa |
| 4 | 4.6 | 3.1 | 1.5 | 0.2 | Setosa |
| 5 | 5.0 | 3.6 | 1.4 | 0.2 | Setosa |
| 6 | 5.4 | 3.9 | 1.7 | 0.4 | Setosa |

- **Step 3: Split the Dataset into Training and Testing Sets**

  We split the dataset into 70% training and 30% testing.

```
set.seed(123)
train_index <- createDataPartition(iris$Species, p=0.7,
                                   list=FALSE)
train_data <- iris[train_index,]
test_data <- iris[-train_index,]
```

- **Step 4: Train the Naïve Bayes Model**

  We build the Naïve Bayes model using the naiveBayes() function from the e1071 package.

```
# Train Naïve Bayes classifier
nb_model <- naiveBayes(Species~., data= train_data)

# Print model details
print(nb_model)
```

- **Step 5: Make Predictions on Test Data**

  We use the trained model to make predictions on the test dataset.

```
# Make predictions
Predictions <- predict(nb_model, test_data)

# Display predictions
print(predictions)
```

```
> print (nb-model)
Naive Bayes Classifier for Discrete predictions
Call:
Naive Bayes.default (x = X, y = Y, laplace = laplace)
A-priori probabilities:
Y
    setosa    versicolor    verginica
 0.3333333    0.3333333    0.3333333
Conditional probabilities:
                sepal.length
Y               [,1]       [,2]
setosa          4.991429   0.3657156
Versicolor      5.942857   0.4558674
Virginica       6.631429   0.6846087
                sepal.width
Y               [,1]       [,2]
setosa          3.365714   0.3455114
versicolor      2.777143   0.3227508
Virginica       2.982857   0.3222035
                petal.length
Y               [,1]       [,2]
setosa          1.471429   0.1808012
versicolor      4.262857   0.4222389
Virginica       5.591429   0.5741344
                Petal.width
Y               [,1]       [,2]
setosa          0.2314286  0.9321521
Versicolor      1.3285714  0.2051848
Virginica       2.0342857  0.2777537
> print (predictions)
[1]  setosa     setosa     setosa     setosa     setosa     setosa
[9]  setosa     setosa     setosa     setosa     setosa     setosa
[17] Virginica  versicolor versicolor versicolor versicolor virginica
[25] versicolor versicolor verginica  versicolor versicolor versicolor
[33] versicolor virginica  verginica  verginica  versicolor Virginica
[41] versicolor versicolor verginica  verginica  verginica
     setosa     versicolor verginica
```

- **Step 6: Evaluate Model Performance**

  We compute accuracy using a confusion matrix.

```
# Confusion matrix
conf_matrix <- confusionMatrix(predictions,
                               test_data$Species)

# Print accuracy
print(conf_matrix$overall['Accuracy'])
```

- **Step 7: Visualize Classification Results**

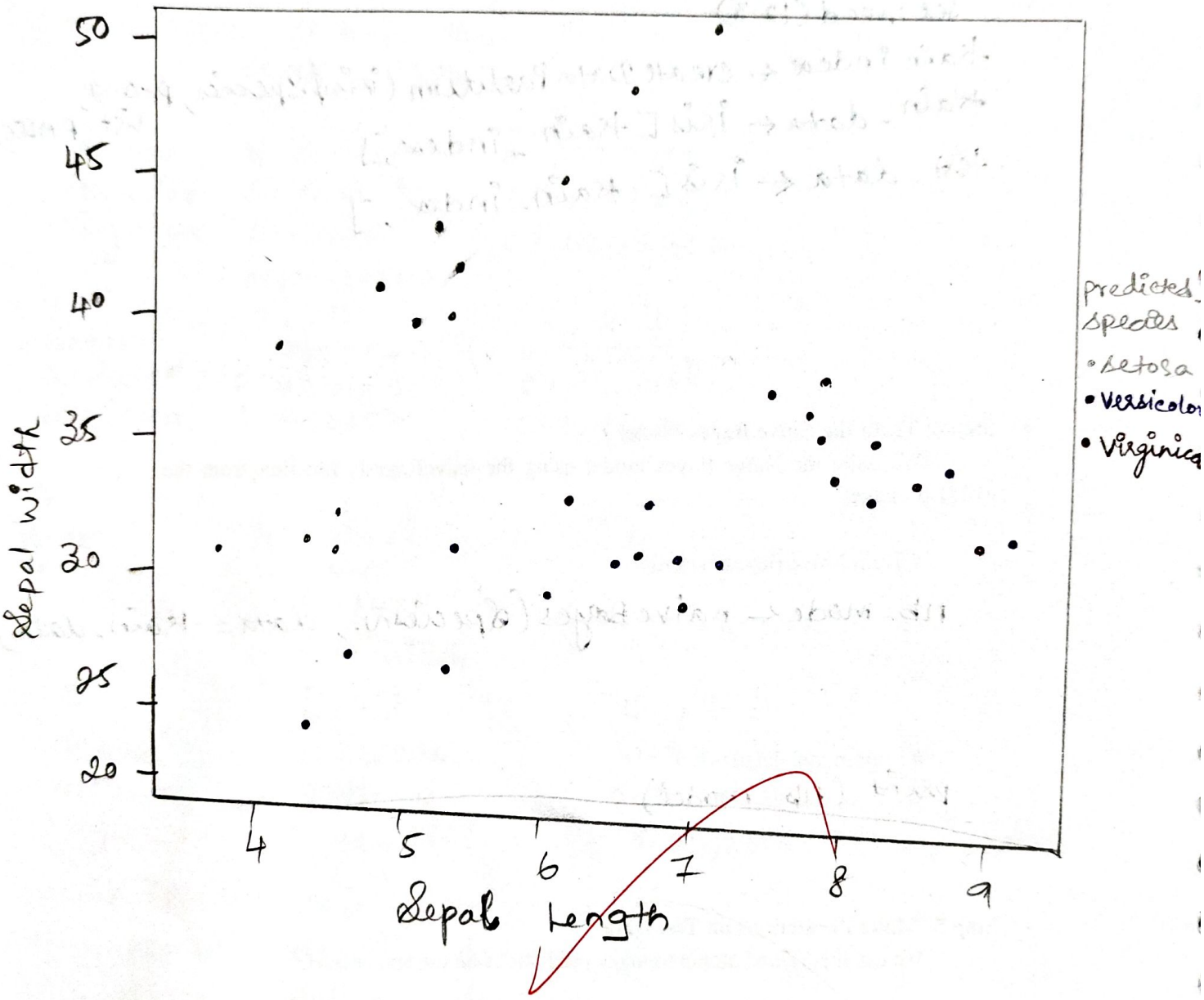
  We plot a scatter plot of Sepal.Length vs Sepal.Width with predicted class labels.

```
# Add predictions to test data
test_data$Predicted_species <- predictions

# Scatter plot of Sepal.Length vs Sepal.Width
ggplot(test_data, aes(x = Sepal.Length, y = Sepal.
Width,
color = Predicted_species)) + geom_point(size = 3) +
ggtitle("Naive Bayes Classification Results")
```

**Result:**

Thus, the program successfully demonstrates the use of the Naive Bayes algorithm in R to classify iris flower species based on numerical attributes. The visualizations clearly show species separation, and the model achieves high classification accuracy, validating the effectiveness of Naive Bayes for multi-class classification problems.

```
> print (conf_matrix $ overall ['Accuracy'])
Accuracy
0.9111111111
```

Naive Bayes Classification Results